

Product Requirements Document

Product Name: Aftermath AI

GenAI-powered Mental Wellness Companion for High-Stakes Exam Students

1. Problem Statement

Students preparing for high-stakes exams such as board exams, NEET, JEE, CUET, CAT, GATE, UPSC, and similar competitive tests experience intense stress, burnout, self-doubt, isolation, and emotional crashes.

Most existing mental wellness tools rely on students voluntarily journaling or logging moods. This breaks down exactly when students are most distressed: after a bad exam, poor score, disappointing mock test, parent conflict, or emotional collapse, they may stop talking, avoid the topic, or disappear entirely.

The key product insight is:

The most important distress signal may not be what the student says. It may be what they avoid, miss, or fail to follow up on after a high-stakes academic event.

For example, if a student expected a high score based on mock tests, took the real exam, then stopped engaging and never reported the result, the system should treat that missing closure as a meaningful signal.

The MVP aims to build a GenAI-powered agent that detects these gaps, gently checks in, understands hidden stress patterns, supports recovery after bad academic events, and allows trusted people such as study buddies, parents, or counsellors to add emergency context when the student cannot self-report.

2. Product Vision

Build an exam-aware emotional support companion that helps students recover before academic stress turns into burnout or crisis.

The product should feel like:

“This app remembers what matters, notices when something may have gone wrong, and checks in without judging me.”

It should not feel like:

“My parents, teachers, or coaching centre are spying on me.”

3. MVP Scope

For the hackathon MVP, the product will focus on one core flow:

MVP Flow: Aftermath Mode + Care Circle Signal

1. Student adds an upcoming exam or mock test.
 2. Student optionally adds expected performance or confidence level.
 3. Exam date passes.
 4. Student does not update the outcome.
 5. System detects missing closure.
 6. Student either:
 7. returns with an unrelated query, or
 8. remains silent, or
 9. receives a concern update from a trusted buddy.
 10. AI gently checks in without asking for marks first.
 11. Student reports that the exam went badly or that they cannot talk.
 12. AI enters Aftermath Mode:
 13. validates the emotional shock,
 14. asks a low-pressure safety check,
 15. offers tap-based response options,
 16. suggests a tiny recovery step,
 17. optionally involves a trusted buddy or counsellor.
 18. If a severe safety concern is detected, the AI stops normal coaching and triggers a human escalation flow.
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4. Target Users

Primary Users

Students preparing for high-pressure academic milestones:

- Class 10 and Class 12 board exam students
- NEET aspirants
- JEE aspirants
- CUET aspirants
- CAT / GMAT / GRE aspirants
- GATE aspirants

- UPSC / government exam aspirants
- Drop-year students
- Coaching institute students

Secondary Users

Trusted support people chosen by the student:

- Study buddy
- Senior
- Classmate
- Sibling
- Close friend
- Girlfriend / boyfriend
- Parent / guardian
- Teacher / mentor
- Counsellor

Institutional Users

For MVP, institutional users are not the core interface, but the product may later support:

- Schools
 - Colleges
 - Coaching institutes
 - Counselling teams
 - Student wellness departments
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5. User Personas

Persona 1: The Silent High Performer

Name: Aarav

Age: 18

Exam: JEE / GMAT / CAT

Behaviour: Scores well in mocks but collapses emotionally after unexpected underperformance.

Pain Point: Does not tell anyone immediately when something goes badly.

Need: A system that notices when a major exam happened and he never closed the loop.

Goal:

"I want someone to check in without making me feel like a failure."

Persona 2: The Burnt-Out Drop-Year Student

Name: Meera

Age: 19

Exam: NEET

Behaviour: Studies long hours, has unstable sleep, feels lonely, and hides emotional exhaustion from parents.

Pain Point: Scores may look stable, but emotional state is deteriorating.

Need: Longitudinal pattern detection from mood, silence, buddy reports, and exam pressure.

Goal:

"I want help before I completely break down."

Persona 3: The Study Buddy

Name: Rohan

Age: 18

Role: Friend / classmate / senior

Behaviour: Notices when the student is crying, isolating, or saying worrying things.

Pain Point: Wants to help but does not know whether to call parents, counsellor, or stay quiet.

Need: A structured way to submit concern without becoming a surveillance tool.

Goal:

"I want to tell the system something is wrong without exposing private drama."

Persona 4: The Concerned Parent

Name: Priya

Role: Parent

Behaviour: Notices mood swings, anger, isolation, and sleep disruption but may misread them as laziness or indiscipline.

Pain Point: Wants visibility but must not violate the student's trust.

Need: Guidance on how to support, not raw access to the student's private journal.

Goal:

"I want to help without making things worse."

Persona 5: The Counsellor / Mentor

Name: Dr. Shah

Role: School or coaching counsellor

Behaviour: Receives cases late, after distress has already escalated.

Pain Point: Lacks structured context about what happened before the crisis.

Need: A concise case summary: triggers, recent events, what helped, what worsened, and current safety status.

Goal:

"I need context quickly so I can support the student safely."

6. User Stories

Student Stories

1. As a student, I want to add my exam dates so that the AI knows when major academic events are happening.
2. As a student, I want to tell the AI my expected score or confidence level so that it understands the emotional weight of the event.
3. As a student, I want the AI to check in after an exam without immediately asking for marks.
4. As a student, I want to respond with simple taps when I do not have energy to type.
5. As a student, I want to say "I can't talk right now" and still feel supported.
6. As a student, I want the AI to remember what makes things worse, such as score pressure or parent comparison.
7. As a student, I want to choose who can support me in emergencies.
8. As a student, I want to control what my parents, buddies, and counsellors can see.
9. As a student, I want the AI to help me recover after a bad exam without shaming me.
10. As a student, I want the AI to help me protect the next paper instead of obsessing over the previous one.

Study Buddy Stories

1. As a study buddy, I want to report that my friend seems severely distressed after an exam.
2. As a study buddy, I want to submit structured concern signals without seeing their private journal.
3. As a study buddy, I want to mark something as urgent if I believe the student may be unsafe.
4. As a study buddy, I want guidance on how to help without making the situation worse.

Parent Stories

1. As a parent, I want guidance on what to say after my child has a bad exam.
2. As a parent, I want emergency alerts only when there is a serious concern.
3. As a parent, I do not need raw private chats; I need support guidance.
4. As a parent, I want to know whether I should give space, stay nearby, or contact a counsellor.

Counsellor Stories

1. As a counsellor, I want a concise escalation summary when a student is flagged.
2. As a counsellor, I want to see recent triggers and concern signals.
3. As a counsellor, I want to update what helped and what worsened after an intervention.
4. As a counsellor, I want the AI to adapt future conversations based on post-escalation context.

7. Functional Requirements

7.1 Student Onboarding

Requirement

The student should be able to create a basic profile for the AI agent.

MVP Fields

- Name or nickname
- Exam track
- Important exam dates
- Preferred tone
- Preferred language
- Optional trusted buddy
- Optional emergency contact
- Consent preferences

Priority

P0

Acceptance Criteria

- Student can enter at least one exam event.
- Student can set at least one trusted support contact.
- Student can choose whether buddy input is allowed.
- Student can choose whether emergency escalation is enabled.

7.2 Exam Timeline Setup

Requirement

Student should be able to add upcoming exams, mocks, or result days.

Fields

- Event name
- Event type: mock, board paper, entrance exam, result day, interview, deadline
- Date and time
- Importance level: low, medium, high
- Expected score or confidence level
- Next dependent event, if any

Priority

P0

Acceptance Criteria

- System stores exam event.
 - System can detect when the event date has passed.
 - System can identify missing outcome after the event.
 - System can trigger a post-event check-in.
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7.3 Mood and Journal Input

Requirement

Student should be able to log emotional state with either quick taps or open-ended journaling.

Input Types

- Mood rating
- Energy rating
- Sleep quality
- Free-text journal
- "I can't talk" option
- One-line vent
- Voice note placeholder for future version

Priority

P0

Acceptance Criteria

- Student can submit a quick mood log in under 30 seconds.
 - Student can submit open-ended text.
 - AI can summarize the emotional signal.
 - AI does not force long journaling.
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7.4 Missing Closure Detection

Requirement

The system should detect when a high-stakes event happened but the student did not report how it went.

Signals

- Exam date passed
- Expected score existed
- Actual score missing
- No post-event mood update
- Student went silent
- Student returned with unrelated query
- Buddy submitted concern

Priority

P0

Acceptance Criteria

- If exam event passes and outcome is missing, system marks event as `outcome_missing`.
 - If student returns after the event without addressing it, AI gently checks in.
 - AI must not ask for marks first.
 - AI must offer low-effort response options.
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7.5 Aftermath Mode

Requirement

The AI should enter a special support mode after a bad exam, missing closure, or concern signal.

Behaviour

The AI should:

- acknowledge the event,
- avoid judgement,
- avoid score-first questioning,
- check emotional state,
- ask if the student is safe if concern is detected,
- offer small coping or grounding steps,
- delay academic planning until student is stable,
- optionally involve buddy or counsellor.

Priority

P0

Acceptance Criteria

- AI recognizes post-exam distress.
 - AI response does not immediately ask for score.
 - AI does not provide harsh productivity advice.
 - AI offers choices such as:
 - "I'm okay"
 - "It went badly"
 - "I can't talk"
 - "I need help"
 - If student indicates severe distress, AI routes to safety workflow.
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7.6 Protect Next Paper Mode

Requirement

If one paper or mock goes badly and another exam is upcoming, the AI should help prevent emotional carryover.

Behaviour

The AI should:

- discourage immediate full post-mortem,
- help the student stabilize,
- identify what is carrying over,
- suggest one small next action,
- recommend rest, food, or support if needed,
- avoid shame-based study planning.

Priority

P1

Acceptance Criteria

- System detects next academic event within a short window.
 - AI offers a "protect next paper" recovery path.
 - AI suggests no more than 1–3 immediate actions.
 - AI avoids long study plans during emotional distress.
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7.7 Care Circle Setup

Requirement

Student should be able to authorize trusted people who can submit concern signals.

Roles

- Study buddy
- Friend / partner
- Parent / guardian
- Mentor / teacher
- Counsellor

Permissions

Each role can have restricted permissions:

- Can submit concern report
- Can receive emergency alert
- Can receive support guidance
- Cannot view private journal by default
- Cannot view private AI chats by default

Priority

P0 for one buddy role

P1 for multiple role types

Acceptance Criteria

- Student can add at least one buddy.
 - Buddy can submit a structured concern report.
 - Buddy cannot access private journal.
 - Student can revoke buddy access.
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7.8 Buddy Concern Report

Requirement

Authorized buddy should be able to submit structured updates.

Fields

- What did you notice?
- When did this happen?

- Did the student say anything specific?
- Did you observe this directly or infer it?
- Is this urgent?
- Is the student safe right now?
- What have you already tried?

Priority

P0

Acceptance Criteria

- Buddy can submit concern without free-form gossip-only input.
- Report distinguishes observation, direct quote, and inference.
- Urgent reports trigger Safety Gate.
- Non-urgent reports are treated as context, not absolute truth.

7.9 Safety Gate

Requirement

The system should detect crisis signals and override normal AI conversation when necessary.

Crisis Signals

- Direct self-harm language
- Suicidal intent
- Plan or means
- Goodbye messages
- Severe hopelessness
- Buddy urgent report
- Student unreachable after urgent concern
- Abuse or immediate danger

Priority

P0

Acceptance Criteria

- Safety check runs before normal AI response.
- Acute concern blocks academic advice.
- AI gives crisis-safe response.
- AI recommends human support.
- AI triggers configured emergency/counsellor path if enabled.
- AI does not claim to diagnose or predict suicide.

7.10 Escalation Event

Requirement

When severe concern is detected, the system should create an escalation event.

Escalation Object

- Student ID
- Trigger signal
- Source
- Timestamp
- Current safety status
- Recommended human contact
- Agent restrictions
- Follow-up required

Priority

P0

Acceptance Criteria

- Escalation record is created.
- AI stops normal productivity/study planning.
- AI switches to safety-first mode.
- Emergency contact/counsellor route is selected according to permissions.

7.11 Post-Escalation Continuity Packet

Requirement

After escalation, the system should record what happened and adapt future conversations.

Continuity Packet Fields

- What triggered escalation
- Current state
- What helped
- What worsened
- What AI should avoid saying
- What AI should prefer saying
- Next vulnerable event
- Follow-up schedule

Priority

P0

Acceptance Criteria

- After escalation, future AI responses use continuity rules.
 - AI does not say judgemental phrases such as “again,” “you promised,” or “why did you fail?”
 - AI does not ask for marks first after crisis.
 - AI begins with safety, steadiness, and consent.
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7.12 Response Critic

Requirement

Before sending sensitive responses, the system should check for harmful wording.

Checks

- Does the response shame the student?
- Does it push productivity during crisis?
- Does it ask for score too early?
- Does it reveal buddy source unnecessarily?
- Does it diagnose?
- Does it ignore safety?
- Does it contradict continuity rules?

Priority

P0

Acceptance Criteria

- Harmful response is blocked or rewritten.
 - AI avoids moral judgement.
 - AI does not reveal third-party source unless appropriate.
 - AI stays within support boundaries.
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7.13 Pattern Summary

Requirement

The AI should summarize possible hidden patterns using cautious language.

Example Output

"Over the last few weeks, your toughest days seem to happen after mock tests where the result feels far below expectation. These days are also followed by silence and avoidance. Does that feel accurate?"

Priority

P1

Acceptance Criteria

- AI presents patterns as hypotheses, not facts.
 - AI asks the student to confirm or correct.
 - AI does not diagnose.
 - AI links pattern to next supportive action.
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8. Non-Functional Requirements

8.1 Safety

- Safety workflow must override all other agent actions.
- The product must not diagnose depression, anxiety, or suicidal risk.
- The product must not claim to replace a therapist, counsellor, doctor, parent, or emergency service.
- The product must route severe distress toward real human support.
- The AI must not generate shame-based, punitive, or comparison-based responses.

8.2 Privacy

- Student owns private journal and AI conversations.
- Buddy, parent, and institution users cannot view private journal by default.
- Third-party reports are source-labelled.
- Student can revoke buddy access.
- Emergency exceptions must be clearly explained during onboarding.
- Sensitive data should be minimized for MVP.

8.3 Consent

- Student must consent before adding care-circle members.
- Student must know what each role can and cannot do.
- Student must be able to edit permissions.
- Third-party input should not become unrestricted surveillance.

8.4 Reliability

- The agent should return structured output for every turn.

- If model confidence is low, system should ask a clarifying question rather than assume.
- If safety state is unknown after a severe signal, the system should prioritize human support.

8.5 Explainability

- Classifier should return evidence for support-state selection.
- Planner should return reason for selected action.
- Escalation event should include trigger source and reasoning.
- Pattern summaries should be explainable to the student.

8.6 Performance

- MVP response time should be under 5 seconds for normal chat.
- Safety responses should be prioritized.
- State retrieval should happen before response generation.

8.7 Accessibility

- Student should be able to respond without typing long text.
- Use simple language.
- Support Hinglish or mixed-language input in future version.
- Avoid clinical jargon in user-facing messages.

9. Acceptance Criteria

MVP Acceptance Criteria

The hackathon MVP is successful if the following demo scenarios work end-to-end.

Scenario 1: Missing Closure After Exam

Given a student had an important exam yesterday
And the student had entered an expected score or confidence level
And the student has not updated the outcome
When the student returns with an unrelated query
Then the AI gently checks in about the unresolved exam
And does not ask for marks first
And offers low-effort response options.

Scenario 2: Bad Exam Aftermath

Given the student says the exam went badly
When the AI responds

Then it validates the shock
And asks how the student is holding up
And checks safety if distress language appears
And does not immediately create a study plan.

Scenario 3: Paper 1 Affects Paper 2

Given Paper 1 went badly
And Paper 2 is tomorrow
When the student says they cannot focus
Then the AI enters Protect Next Paper Mode
And suggests a tiny recovery action
And avoids full post-mortem before Paper 2.

Scenario 4: Buddy Emergency Report

Given an authorized buddy reports that the student said something self-harm related
When the report is submitted
Then the Safety Gate triggers
And normal academic advice is blocked
And an escalation event is created
And the AI prioritizes human support.

Scenario 5: Post-Escalation Continuity

Given a student had an escalation yesterday
And the continuity packet says score discussion worsened distress
When the student returns today
Then the AI does not ask for marks first
And does not say “again” or blame the student
And begins with a gentle check-in.

Scenario 6: Parent Judgement Report

Given a parent reports “my child is lazy and wasting time”
When the system processes the report
Then it does not treat the judgement as fact
And asks for observable behaviour
And does not escalate unless safety evidence exists.

10. Success Metrics

Demo Success Metrics

- Agent correctly identifies missing closure after exam.
- Agent correctly enters Aftermath Mode.
- Agent correctly handles buddy concern input.
- Agent correctly blocks unsafe productivity advice during crisis.
- Agent correctly uses post-escalation continuity memory.
- Judges can understand the product differentiation within 2 minutes.

Product Metrics

Engagement

- Percentage of students who add at least one exam event.
- Percentage of students who complete post-exam check-in.
- Percentage of students using tap-based check-ins.
- Weekly active users during exam period.

Support Quality

- Percentage of AI responses rated “felt supportive.”
- Percentage of students who say “AI understood my situation.”
- Percentage of students who complete a recovery action after bad exam.
- Percentage of students who use Aftermath Mode after exam events.

Safety

- Number of severe concern cases correctly routed to human support.
- Number of unsafe responses caught by response critic.
- False escalation rate.
- Missed crisis signal rate in test simulations.
- Percentage of escalations with completed continuity packet.

Trust and Privacy

- Percentage of students who add a care-circle member.
- Percentage of students who keep buddy permissions enabled after first week.
- Number of permission revocations.
- Student trust rating after buddy/care-circle interaction.

Hackathon MVP Metrics

- End-to-end demo flow completion.
- Latency under 5 seconds for standard response.
- Safety Gate correctly overrides normal chat in test cases.
- At least 5 simulated student journeys pass acceptance tests.

11. Assumptions

1. Students are more likely to respond honestly if the AI does not ask for marks first.
 2. Silence after a high-stakes event can be a meaningful support signal, but it is not proof of crisis.
 3. Trusted buddies may notice distress before the student self-reports.
 4. Third-party reports must be treated as signals, not verified truth.
 5. The MVP will use simulated data rather than real institution integrations.
 6. The MVP will not make medical diagnoses.
 7. The MVP will use support states rather than clinical labels.
 8. The AI agent will be evaluated through scripted scenarios.
 9. Emergency escalation will be simulated or routed to placeholder contacts in the hackathon version.
 10. Full legal, clinical, and institutional review will be required before real deployment.
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12. Out-of-Scope Features for Hackathon MVP

The following are intentionally excluded from MVP:

- Full production mobile app
 - Real-time institution integrations
 - Real coaching platform score sync
 - Payment system
 - Production emergency calling
 - Therapist marketplace
 - Diagnosis of depression, anxiety, PTSD, or suicidality
 - Predictive suicide-risk scoring
 - Long-term clinical treatment plans
 - Full parent dashboard
 - Full institution dashboard
 - Wearable integration
 - Voice emotion detection
 - Social feed or peer community
 - Anonymous public forums
 - Medication advice
 - Therapy replacement claims
 - Fully automated crisis handling without humans
 - Fine-tuned custom ML model
 - Large-scale longitudinal analytics
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13. MVP Technical Scope

The MVP should implement the following AI-agent modules:

1. Signal Normalizer

2. Student State Store
 3. Exam Event Graph
 4. Missing Closure Detector
 5. Safety Gate
 6. Support-State Classifier
 7. Agent Planner
 8. Response Generator
 9. Response Critic
 10. Escalation Event Creator
 11. Continuity Packet Updater
 12. Simulation Test Harness
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14. Support States

The agent should classify users into support states, not diagnoses.

Phase 0: Stable

Student is functioning normally.

Phase 1: Elevated Stress

Student is stressed but still functioning.

Phase 2: Aftermath / Functional Disruption

Student may be affected by a bad exam, missing closure, avoidance, silence, or buddy concern.

Phase 3: Acute Safety Concern

Student may need immediate human support due to self-harm, suicidal language, severe distress, or urgent third-party report.

15. Key Differentiator

Most AI wellness apps respond to what the student says.

This product responds to:

- what the student says,
- what happened academically,
- what was expected to happen,
- what was not said,
- who else is worried,

- what changed afterward,
- what harmed or helped last time.

The product's intelligence lives in the gap between expected and observed behaviour.

16. Final MVP Definition

The hackathon MVP is:

A GenAI-powered exam wellness agent that detects missing emotional closure after high-stakes academic events, gently checks in, supports bad-result recovery, accepts structured concern reports from trusted buddies, escalates severe concerns to human support, and remembers post-escalation context so future conversations remain safe and empathetic.